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Nonstationary Irreversible Thermodynamics: A Causal Relativistic Theory*

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Conventional theories of irreversible thermodynamics predict instantaneous propagation of thermal and viscous effects. This defect can be traced to the omission of terms quadratic in the heat flux and viscous stresses from the expression for the entropy 4-vector in terms of the fluxes of 4-momentum and material. By a systematic restoration of the quadratic terms, a generalized theory is formulated which is applicable to the description of transient irreversible effects and reconcilable with causality. Also discussed is the relativistic formulation of the second law of thermodynamics for the exchange of heat and matter between bodies with arbitrary relative velocities.

1. INTRODUCTION

Current formulations of relativistic irreversible thermodynamics [1–5] contain a fundamental weakness which has disturbed many people [6–9]. They lead to parabolic differential equations, hence to infinite speeds of propagation for heat flow and viscosity, in contradiction with the principle of causality.

While such paradoxes appear particularly glaring in a relativistic theory, infinite propagation speeds already constitute a difficulty at the classical level, since one does not expect thermal disturbances to be carried faster than some (suitably defined) mean molecular speed. Conventional theory is thus applicable only to phenomena which are "quasi-stationary," i.e., slowly varying on space and time scales characterized by the mean free path and mean collision time. This is inadequate for many phenomena in high-energy astrophysics involving steep gradients or rapid variations.

These deficiencies can be traced to the fact that the conventional theories (both classical and relativistic) make overly restrictive hypotheses concerning the relations between the fluxes and densities of entropy, energy, and particle number.

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In the case of a simple fluid, for instance, the conventional theories postulate that entropy flux is simply proportional to heat flux and that entropy density is independent of heat flow and viscosity [10]. This postulate is indeed correct to *first* order in deviations from equilibrium; however, in order to infer the correct linear phenomenological laws from the expression for the entropy production, the latter is required to *second* order. Neglect of second-order terms in the entropy density and flux leads to errors of the same order in the entropy production, except in circumstances where space-time gradients of heat flux and viscous stress are negligible, i.e., under quasistationary conditions.

It is the purpose of this paper to remedy these shortcomings by a systematic retention of second-order terms in the entropy flow 4-vector. This will result in a generalized irreversible thermodynamics for simple fluids and fluid mixtures which is applicable to nonstationary processes and reconcilable with causality.

In the case of a simple fluid, isotropy limits the number of second-order terms in the entropy flow to five. Correspondingly, five new transport coefficients appear in the phenomenological laws, in addition to the familiar thermal conductivity and the two coefficients of viscosity. Three of the new coefficients may be identified as relaxation times for the viscous stresses and heat flow, and the remaining two as coupling coefficients for cross effects between them. (A procedure sometimes used to secure a finite velocity for heat is to add an ad hoc relaxation term to Fourier's law [6, 7]. The present theory amends and rationalizes this procedure.) If the relaxation times are taken to be of the order of a mean collision time (a reasonable assumption), the speeds of thermal and viscous effects will be of the order of the speed of sound.

For the benefit of the reader desiring a quick overview, the principal formulas for simple fluids are assembled (without derivation) in Section 2.

The general theory for fluid mixtures is then systematically developed, beginning in Section 3. The basic ingredients of the theory are: For each component A, a 4-vector $N_A{}^\mu$ representing the numerical flux of particles of type A, and, for the mixture as a whole, the energy-momentum tensor $T^{\lambda\mu}$ (symmetric and conserved) and the entropy flux S^μ (with positive semi-definite divergence). To make further progress, one must assume that deviations from local equilibrium (i.e., the heat flux and viscous stresses) are small. In that case there exists a relation which links S^μ, $T^{\lambda\mu}$, and $\{N_A{}^\mu\}$, and which is the fundamental equation of the theory. This is discussed in Section 6. In Section 7 we obtain the entropy production and deduce the linear phenomenological laws for a mixture.

The theory just described applies to a mixture of strongly interacting components close to a common equilibrium, so that, in particular, the relative velocities of the components are small. Section 8 contains a brief sketch of a quite different formalism applicable to a system of weakly interacting constituents in which each constituent is close to a state of local individual equilibrium, but the system

as a whole may be arbitrarily far from equilibrium. This permits the constituents to have arbitrary relative velocities and provides a novel approach to the relativistic second law of thermodynamics for heat exchange between rapidly moving bodies, a subject which has generated much heat and some entropy in the last decade.

It is important to mention that a relativistic kinetic theory based on the Boltzmann equation can be developed [11] along lines entirely parallel to the present phenomenological theory. This confirms the correctness of our treatment in the special case of gases, and also provides explicit expressions for the new transport coefficients.

2. Resumé of Results for a Simple Fluid

An arbitrary (nonequilibrium) state of a simple fluid is described by the energy tensor $T^{\lambda\mu}$ (symmetric and conserved), a conserved particle flux vector N^μ and an entropy flux S^μ satisfying the second law of thermodynamics: i.e.,

$$T^{\lambda\mu}{}_{|\mu} = 0, \qquad N^\mu{}_{|\mu} = 0, \qquad S^\mu{}_{|\mu} \geqslant 0, \tag{1}$$

in which the stroke indicates covariant differentiation.

We shall assume that the energy density is nonnegative, i.e., $T^{\lambda\mu}u_\lambda u_\mu \geqslant 0$ for all timelike u_μ. A theorem due to Synge [14] then shows that $T^{\lambda\mu}$ has a unique timelike eigenvector $u_E{}^\mu$, normalized by $u_E{}^\mu u_{E\mu} = -1$. Let $u_N{}^\mu$ be the unit timelike vector parallel to N^μ.

In equilibrium, S^μ, $u_E{}^\mu$, and $u_N{}^\mu$ are all parallel, defining a unique "hydrodynamical 4-velocity" and "rest-frame."

Away from equilibrium this is no longer true. For small deviations we have, for instance,

$$u_E{}^\mu = u_N{}^\mu + (\rho + P)^{-1} q^\mu + O_2 . \tag{2}$$

where q^μ is the heat flux vector, ρ and P the energy density and thermodynamical pressure (we use relativistic units with $c = 1$). The symbols O_1, O_2 will be used throughout to denote unspecified terms of first and second order in the deviations from local equilibrium (i.e., in the heat flux and viscous stresses). Thus, for nonequilibrium processes the choice of a "hydrodynamical velocity" u_μ is ambiguous, and, indeed, such a choice has no physical significance, since the basic physical relations (1) and the primary variables $T^{\lambda\mu}$, N^μ, and S^μ make no reference to any "rest-frame." This point will be made explicit in the following sections, where we shall develop the thermodynamics of small deviations from equilibrium in a form manifestly invariant under first-order changes of rest-frame, so that any hydrodynamical velocity u^μ which differs from $u_E{}^\mu$ or $u_N{}^\mu$ by O_1 will be admissible [3].

Of course, gradients of u_μ do appear explicitly in the phenomenological stress/rate-of-strain relations we shall ultimately derive, so that, in applying these equations some definite (though arbitrary) choice of u_μ must be made. Matters are very simple in the quasi-stationary case, where it turns out that the phenomenological equations are insensitive to first-order changes of u_μ. This is because (i) quite generally, an O_1 change in u_μ produces only an O_2 change in thermodynamical parameters such as the energy density $\rho = u_\alpha u_\rho T^{\alpha\beta}$ and particle density $n = -u_\alpha N^\alpha$ measured in the frame u_μ; and (ii) under quasi-stationary conditions we can neglect the difference between gradients of $u_E{}^\mu$ and $u_N{}^\mu$, which is essentially the gradient of heat flux (see (2)). For nonstationary processes (ii) is not valid, and (i) requires caution when considering the entropy flux, since that must be evaluated to second order.

In Section 7, the phenomenological equations will be expressed in a form independent of the choice of rest-frame. Here, it may be helpful to exhibit their specific forms for the two simplest choices, $u_\bullet^\mu = u_E{}^\mu$ and $u^\mu = u_N{}^\mu$.

(a) *Energy (or E-) Frame* ($u^\mu = u_E{}^\mu$)

In this frame, used by Landau and Lifshitz [2] in their quasi-stationary analysis, the energy tensor and the Einstein field equations reduce to their simplest form. To an observer at rest in the energy frame, the flux of energy appears to be zero. A "heat flux" (i.e., flow of energy relative to particle stream) along the positive x-axis, say, appears as a drift of particles towards negative x-values.

With respect to $u_E{}^\mu$, the numerical flux and energy tensor decompose as

$$N^\alpha = n u_E{}^\alpha - \eta^{-1} q^\alpha \tag{3}$$

$$T^{\alpha\beta} = \rho u_E{}^\alpha u_E{}^\beta + (P + \sigma)\, \Delta_E^{\alpha\beta} + \sigma^{\alpha\beta}. \tag{4}$$

Here, $\Delta_E^{\alpha\beta} = \Delta^{\alpha\beta}(u_E)$, and

$$\Delta^{\alpha\beta}(u) = g^{\alpha\beta} + u^\alpha u^\beta \tag{5}$$

is defined as the operator which projects onto the 3-space orthogonal to an arbitrary unit timelike vector u^α. The quantity

$$\eta = (\rho + P)/n \tag{6}$$

is the relativistic enthalpy or "injection energy" [12] per particle, and q^α, σ, and $\sigma^{\alpha\beta}$ denote heat flux, bulk and shear stress; they satisfy

$$q^\alpha u_\alpha = u_\alpha \sigma^{\alpha\beta} = 0, \qquad \sigma^{\alpha\beta} = \sigma^{\beta\alpha}, \qquad \sigma_\alpha{}^\alpha = 0. \tag{7}$$

In terms of gradients of $u_E{}^\alpha$, the linear phenomenological laws take the form

$$\sigma = -\tfrac{1}{3}\zeta_V(u^\alpha_{E|\alpha} + \beta_0\dot{\sigma} - \alpha_0 q^\alpha_{|\alpha}) \tag{8a}$$

$$q^\alpha = \kappa T \Delta^{\alpha\beta}_E(\eta^{-1}T\Theta_{,\beta} - \beta_1\dot{q}_\beta + \alpha_0\sigma_{,\beta} + \alpha_1\sigma^\gamma_{\beta|\gamma}) \tag{8b}$$

$$\sigma_{\alpha\beta} = -2\zeta\langle u^E_{\alpha|\beta} + \beta_2\dot{\sigma}_{\alpha\beta} - \alpha_1 q_{\alpha|\beta}\rangle \tag{8c}$$

Beside the three familiar transport coefficients ζ_V and ζ (bulk and shear viscosity) and κ (thermal conductivity), these equations contain five new phenomenological coefficients: three relaxation times proportional to β_0, β_1, β_2, and two parameters α_0, α_1 which couple heat flow and viscous stresses. The "thermal potential" Θ is defined as (relativistic chemical potential per particle)/(temperature), i.e.,

$$\Theta = \frac{(\rho + P - TS)}{nT} \tag{9}$$

$$= \frac{m}{T}\left[1 + \frac{(\text{classical chemical potential})}{c^2}\right] \text{(c.g.s. units)}; \tag{10}$$

T is temperature, S entropy density, and m the rest-mass per particle. The angular brackets in (8c) denote the symmetric, purely spatial, trace-free part of the enclosed tensor, i.e.,

$$\langle A_{\alpha\beta}\rangle \equiv \tfrac{1}{2}\Delta_\alpha{}^\lambda\Delta_\beta{}^\mu(A_{\lambda\mu} + A_{\mu\lambda} - \tfrac{2}{3}\Delta_{\lambda\mu}\Delta^{\rho\sigma}A_{\rho\sigma}). \tag{11}$$

Partial derivatives in (8) are denoted by a comma. A dot denotes the proper-time derivative, e.g., $\dot{q}_\beta = q_{\beta|\mu}u^\mu$.

With α_i and β_i set equal to zero, Eqs. (8) reduce to the quasi-stationary equations given by Landau and Lifshitz [2].

(b) *Particle Frame or N-Frame* ($u^\mu = u_N{}^\mu$)

The decompositions of N^α and $T^{\alpha\beta}$ now take the form

$$N^\alpha = nu_N{}^\alpha \tag{12}$$

$$T^{\alpha\beta} = \rho u_N{}^\alpha u_N{}^\beta + (P + \sigma)\,\Delta^{\alpha\beta}_N + q^\alpha u_N{}^\beta + q^\beta u_N{}^\alpha + \sigma^{\alpha\beta} \tag{13}$$

where $\Delta^{\alpha\beta}_N = \Delta^{\alpha\beta}(u_N)$.

The form (12) permits easy integration of the conservation law $N^\alpha{}_{|\alpha} = 0$, and this makes the N-frame advantageous for some special-relativistic problems (e.g., the analysis of relativistic shock structure [3, 13]. Also, it is slightly more "intuitive" than the E-frame, which is perhaps why Eckart [1] and many subsequent authors have chosen it.

Referred to gradients of $u_N{}^\alpha$, the phenomenological laws are

$$\sigma = -\tfrac{1}{3}\zeta_V(u^\alpha_{N|\alpha} + \beta_0\dot{\sigma} - \bar{\alpha}_0 q^\alpha_{|\alpha}) \tag{14a}$$

$$q^\alpha = -\kappa T\Delta^{\alpha\beta}_N(T^{-1}T_{,\beta} + \dot{u}_\beta{}^N + \bar{\beta}_1\dot{q}_\beta - \bar{\alpha}_0\sigma_{,\beta} - \bar{\alpha}_1\sigma^\gamma_{\beta|\gamma}) \tag{14b}$$

$$\sigma_{\alpha\beta} = -2\zeta\langle u^N_{\alpha\beta} + \beta_2\dot{\sigma}_{\alpha\beta} - \bar{\alpha}_1 q_{\alpha|\beta}\rangle. \tag{14c}$$

The parameters $\bar{\alpha}_0$, $\bar{\alpha}_1$, $\bar{\beta}_1$ are related to the corresponding unbarred parameters in (8) by

$$\bar{\alpha}_i - \alpha_i = \beta_1 - \bar{\beta}_1 = [(\rho + P)\,T]^{-1}. \tag{15}$$

Apart from these, quantities bearing the same symbol in (8) and (14) have the same meaning and value, at least to within the O_1 accuracy of the linear laws.

The equivalence of the sets (8) and (14) can be established with the aid of (2) and the result

$$\Delta^{\alpha\beta}\eta^{-1}\Theta_{,\beta} = \Delta^{\alpha\beta}\left\{\left(\frac{1}{T}\right)_{,\beta} - \frac{1}{(\rho + P)T}(\dot{q}_\beta + \sigma_{,\beta} + \sigma^\gamma_{\beta|\gamma})\right\} - \frac{1}{T}\,\dot{u}_N{}^\alpha, \tag{16}$$

which follows from the thermodynamical identity

$$\eta^{-1}\,d\Theta = d(T^{-1}) + [(\rho + P)\,T]^{-1}\,dP, \tag{17}$$

and the spatial part of the conservation identity $T^{\alpha\beta}{}_{|\beta} = 0$, evaluated in the N-frame. The equivalence follows more easily from the general formulas of Section 7.

Because the phenomenological laws (8) and (14) are in any case valid to accuracy O_1 only, we have often allowed our notation to blur distinctions which are unimportant at this level. For example, $\rho_E = u_E{}^\alpha u_E{}^\beta T_{\alpha\beta}$ and $\rho_N = u_N{}^\alpha u_N{}^\beta T_{\alpha\beta}$, which differ by O_2, have been written simply as ρ, and for similar reasons it is immaterial whether u^α in (6) and $\Delta^{\alpha\beta}$ in (16) have E or N values. More care must now be exercised as we record expressions for the entropy flux *accurate to* O_2.

In the E-frame,

$$S^\mu = S_E u_E{}^\mu + \eta^{-1}\Theta q_E{}^\mu - Q_E{}^\mu, \tag{18}$$

where $Q_E{}^\mu$ represents the O_2 contribution

$$TQ_E{}^\mu = \tfrac{1}{2}u_E{}^\mu(\beta_0\sigma^2 + \beta_1 q^\alpha q_\alpha + \beta_2\sigma^{\alpha\beta}\sigma_{\alpha\beta}) - \alpha_0\sigma q^\mu - \alpha_1\sigma_\nu{}^\mu q^\nu. \tag{19}$$

In the N-frame,

$$S^\mu = S_N u_N{}^\mu + T^{-1}q_N{}^\mu - Q_N{}^\mu; \tag{20}$$

$Q_N{}^\mu$ is given by the expression (19) with τ_1, α_0, α_1 replaced by the corresponding barred quantities (15). The quantities S_E, S_N are defined by

$$S_E = S_{\text{eq}}(\rho_E, n_E), \qquad S_N = S_{\text{eq}}(\rho_N, n_N), \tag{21}$$

where $S = S_{\text{eq}}(\rho, n)$ is the *equilibrium* equation of state. The vectors

$$q_E{}^\mu = -\eta_E \Delta_E^{\mu\nu} N_\nu, \qquad q_N{}^\mu = -\Delta_N^{\mu\alpha} u_N{}^\beta T_{\alpha\beta} \tag{22}$$

are the q's entering (3) and (13), respectively. The differences $q_E{}^\mu - q_N{}^\mu$ and $S_E - S_N$ are of O_2.

The divergence of (18) or (20), together with the phenomenological laws, yields the positive definite entropy production

$$S^\mu{}_{|\mu} = \frac{q^\mu q_\mu}{\kappa T^2} + \frac{3\sigma^2}{\zeta_V T} + \frac{\sigma^{\lambda\mu}\sigma_{\lambda\mu}}{2\zeta T}. \tag{23}$$

3. General Relations for Fluid Mixtures

The primary variables determining a general (nonequilibrium) macroscopic state of a fluid mixture will be taken (following Aitken [3]) to be the energy tensor $T^{\alpha\beta}$, the entropy flux S^μ and a set of timelike vectors N^α ($A = 1, 2, \ldots$) giving the numerical flux of particles of component A.

For an isolated system,

$$T^{\alpha\beta}{}_{|\beta} = 0, \qquad S^\mu{}_{|\mu} \geqslant 0 \tag{24}$$

express conservation of energy–momentum and the law of increase of entropy. If there are chemical reactions, the rate of production of species A per unit volume will be given by

$$N_{A|\alpha}^\alpha = \sum_i c_{Ai} r_i, \tag{25}$$

where r_i is the "rate" of the ith reaction, and c_{Ai} the stoichiometric coefficients for this reaction. (Example: for the reaction $A + B \to C + 3B$, $c_A = -1$, $c_B = 3 - 1$, $c_C = +1$.) If e_A, b_A, ... are the charge, baryon number,... of a particle of species A, then

$$\sum_A e_A c_{Ai} = \sum_A b_A c_{Ai} = \cdots = 0 \tag{26}$$

are the conditions that these quantities are conserved in the ith reaction.

4. Equilibrium

An equilibrium state of the system is defined by the following properties.

(i) The entropy production vanishes:

$$S^{\mu}{}_{|\mu} = 0. \tag{27}$$

(ii) All primary variables are expressible in terms of a unique "hydrodynamical 4-velocity" u^{μ}:

$$S^{\mu} = Su^{\mu}, \qquad N_A{}^{\mu} = n_A u^{\mu}, \qquad T^{\lambda\mu} = \rho u^{\lambda} u^{\mu} + P\Delta^{\lambda\mu}. \tag{28}$$

(The projection tensor $\Delta^{\lambda\mu}(u)$ was defined in (5).) Equations (28) identify S, ρ, P, and n_A as the entropy density, energy density, pressure, and particle density for species A, measured in the hydrodynamical rest-frame.

(iii) There is a functional relation ("equilibrium equation of state")

$$S = S(\rho, n_A). \tag{29}$$

Then

$$dS = T^{-1}\, d\rho - \sum_A \Theta_A\, dn_A \tag{30}$$

defines inverse temperature T^{-1} and "thermal potentials" Θ_A ($= T^{-1} \times$ relativistic chemical potentials, compare Eq. (10)) as partial derivatives of the function $S(\rho, n_A)$.

(iv) The thermodynamical pressure $P(\rho, n_A)$ is given by

$$S = T^{-1}(\rho + P) - \sum_A \Theta_A n_A\,. \tag{31}$$

(v) The thermal potentials satisfy the "chemical equilibrium relations"

$$\sum_A \Theta_A c_{Ai} = 0 \qquad (i = 1, 2, \ldots). \tag{32}$$

(vi) The reaction rates r_i vanish.

(vii) The motion is rigid in the sense of Born [14]:

$$\Delta^{\lambda\alpha}\Delta^{\mu\beta}(u_{\alpha|\beta} + u_{\beta|\alpha}) = 0. \tag{33}$$

This requires the orthogonal distance between adjacent material world-lines to stay constant, so that viscous forces do not arise.

(viii) Finally, the conditions that heat flow and diffusion vanish are

$$\Theta_A = \text{const.} \tag{34}$$

We append a few comments on these conditions.

Conditions (iv) and (v) can be plausibly "derived" from the postulates preceding them by the following reasoning. A "quasi-static process," in which the system passes through a sequence of equilibrium states very slowly (expansion rate $u^\mu{}_{|\mu}$ and reaction rates r_i very small), should be reversible (i.e., the entropy production $S^\mu{}_{|\mu}$ should vanish to first order). Moreover, we may assume that Eqs. (28) remain valid when computing $S^\mu{}_{|\mu}$ to first order, since the effect of transport terms on $S^\mu{}_{|\mu}$ is of second order. Computing $S^\mu{}_{|\mu}$ with the aid of (28), (30), (25) and $T^{\alpha\beta}{}_{|\beta} = 0$ then yields, to first order

$$S^\mu{}_{|\mu} = \left[S - T^{-1}(\rho + P) + \sum_A \Theta_A n_A\right] u^\mu{}_{|\mu} - \sum_{A,i} \Theta_A c_{Ai} r_i = 0. \tag{35}$$

This leads to (31) and (32) if the rates $u^\mu{}_{|\mu}$ and r_i are considered independent.

Equations (30) and (31) imply the important thermodynamical identity

$$T^{-1}\, dP + (\rho + P)\, d(T^{-1}) = \sum_A n_A\, d\Theta_A \tag{36}$$

and also the result

$$T\, d(SV) = d(\rho V) + P\, dV - T \sum_A \Theta_A\, d(n_A V) \tag{37}$$

which is valid for an arbitrary parameter V. It resembles the most familiar form of the classical Gibbs relation if V is considered to be some kind of specific volume; for example, $V^{-1} = \sum_A n_A m_A$ where m_A is the rest-mass of a particle of species A (assuming that not all m_A vanish), or $V^{-1} = \sum_A n_A b_A$, where b_A is baryon number.

From (33),

$$u^\mu{}_{|\mu} = 0. \tag{38}$$

Conditions (vi) and (i) then imply, with the aid of Eqs. (25), (28) and (30), that

$$\dot{n}_A = \dot{\rho} = 0, \tag{39}$$

where the dot denotes the proper-time derivative $u^\mu \partial_\mu$. Thus, all thermodynamical parameters are independent of time.

The conservation law $T^{\lambda\mu}{}_{|\mu} = 0$ gives for the acceleration $\dot{u}_\lambda = u_{\lambda|\mu} u^\mu$,

$$(\rho + P)\, \dot{u}_\lambda = -\partial_\lambda P, \tag{40}$$

where we have used the fact that $\dot{P} = 0$. Equation (33) then leads to the expression

$$(\rho + P)\, u_{(\lambda|\mu)} = u_{(\lambda}\partial_{\mu)}P, \tag{41}$$

in which parentheses enclosing indices indicate symmetrization. By invoking (34) and (36) we conclude that

$$(T^{-1}u_{(\lambda)|\mu)} = 0, \tag{42}$$

so that $\xi_\lambda \equiv T^{-1}u_\lambda$ is a Killing vector field, and the gravitational field is stationary. In particular, we can deduce the well-known Tolman condition

$$T(-\xi_\lambda\xi^\lambda)^{1/2} = \text{const.} \tag{43}$$

for thermal equilibrium in a gravitational field, as well as the Tolman–Klein condition [15]

$$(T\Theta_A)(-\xi_\lambda\xi^\lambda)^{1/2} = \text{const.} \tag{44}$$

for the chemical potentials $(T\Theta_A)$. In special relativity, (42) leads to the law

$$T(1 - v^2/c^2)^{1/2} = \text{const.} \tag{45}$$

for the equilibrium temperature distribution in a rigidly spinning body. The hottest parts of the body are at the farthest distance from the axis, an effect which can be attributed to the "inertia of heat."

5. Off-Equilibrium

We return to a general nonequilibrium state described by the primary variables $N_A{}^\mu$, $T^{\lambda\mu}$, S^μ.

We now assume that deviations of the components from a state of common local equilibrium are small. These deviations are the dimensionless ratios

$$(\nu_A{}^\mu/n_A\,,\ \tau^{\lambda\mu}/\rho) = O_1 \tag{46}$$

where $\tau^{\lambda\mu}$ are the viscous stresses and the flux $\nu_A{}^\mu$ measures the drift of component A relative to the energy frame. (Precise definitions will be given presently.) As in Section 2, the symbols $O_1, O_2, \ldots$ will denote quantities of (at least) first, second,... order in the deviations. Let $u_A{}^\mu$ be the unit timelike vector parallel to $N_A{}^\mu$ and $u_E{}^\mu$ the unit time-like eigenvector of $T^{\lambda\mu}$. Then the condition $\nu_A{}^\mu/n_A = O_1$ is equivalent to

$$u_A{}^\mu - u_E{}^\mu = O_1\,, \tag{47}$$

i.e., the (pseudo-) angles between the various 4-velocities are small of first order.

Choose any unit timelike vector u^μ such that

$$u^\mu - u_E{}^\mu \leqslant O_1 . \tag{48}$$

The energy density $\rho(u)$ and particle densities $n_A(u)$ measured by an observer having this 4-velocity are given by

$$\rho(u) = u_\alpha u_\beta T^{\alpha\beta}, \qquad n_A(u) = -u_\alpha N_A{}^\alpha. \tag{49}$$

Define the quantity $S(u)$ by

$$S(u) = S(\rho(u), n_A(u)), \tag{50}$$

where the function $S(\rho, n_A)$ is the one that occurs in the *equilibrium* equation of state (29). By means of (30) and (31) we can then define $T(u)$, $\Theta_A(u)$ and $P(u)$ in terms of partial derivatives of S. The thermodynamical identities (36) and (37) remain valid for the quantities thus defined.

(It should be emphasized at once that (50) is purely a formal definition; $S(u)$ is not "the" entropy. The true entropy is entirely determined by the primary variable S^μ, which has not yet entered the scene. However, it will turn out (see Eq. (65) below) that $S(u)$ is an approximation to the entropy density, accurate to first order. This is all that is needed for our purpose, which is to secure a correct first-order separation between (a) the thermodynamical pressure P, defined by (31), or equivalently by

$$P = -[\partial(\rho V)/\partial V]_{(SV, n_A V)=\text{const.}} \tag{51}$$

as the work done in a *reversible* expansion, and (b) the irreversible part σ of the total bulk stress. Of the variables T, Θ_A, and P, only the last has more than a formal significance, since it enters directly into the expression for the energy tensor.)

As seen in the frame with 4-velocity u^α, the energy flow and particle drift for species A are given by the purely spatial vectors

$$h^\alpha(u) = -\Delta_\lambda{}^\alpha(u)\, u_\mu T^{\lambda\mu}, \qquad j_A{}^\alpha(u) = \Delta_\lambda{}^\alpha(u)\, N_A{}^\lambda, \tag{52}$$

and the viscous stress by

$$\tau^{\alpha\beta}(u) = \Delta_\lambda{}^\alpha(u)\, \Delta_\mu{}^\beta(u)\, T^{\lambda\mu} - P(u)\, \Delta^{\alpha\beta}(u). \tag{53}$$

This can split into a pure shear and a bulk stress:

$$\tau^{\alpha\beta}(u) = \sigma^{\alpha\beta}(u) + \sigma(u)\, \Delta^{\alpha\beta}(u), \qquad \sigma_\alpha{}^\alpha = 0. \tag{54}$$

These definitions provide a unique decomposition of the primary variables N^{α}, $T^{\alpha\beta}$ with respect to the arbitrary 4-velocity u^{α}:

$$N_A{}^{\alpha} = n_A(u)\, u^{\alpha} + j_A{}^{\alpha}(u), \tag{55}$$

$$T^{\alpha\beta} = \rho(u)\, u^{\alpha}u^{\beta} + P(u)\, \Delta^{\alpha\beta}(u) + 2u^{(\alpha}h^{\beta)}(u) + \tau^{\alpha\beta}(u), \tag{56}$$

$$j_A{}^{\alpha}u_{\alpha} = h^{\alpha}u_{\alpha} = u_{\alpha}\tau^{\alpha\beta} = 0. \tag{57}$$

All these quantities, with the exception of the primary variables, are frame-dependent. In the Appendix, we examine how they are affected by the change of frame $u^{\alpha} \to \bar{u}^{\alpha}$, where

$$\bar{u}^{\alpha} = u^{\alpha}(1 + \epsilon^2)^{1/2} + \epsilon^{\alpha}, \qquad \epsilon^{\alpha}u_{\alpha} = 0, \quad \epsilon^2 \equiv \epsilon^{\alpha}\epsilon_{\alpha}\,. \tag{58}$$

To maintain (48), we must assume $\epsilon \leqslant O_1$. The results are as follows.

(i) The viscous stresses $\tau^{\alpha\beta}$ and all thermodynamic parameters ρ, n_A, P, S, T, and Θ_A change by amounts of order $\epsilon O_1 \leqslant O_2$. They may be considered effectively frame-invariant insofar as their appearance in the linear phenomenological laws is concerned.

(ii) The particle drift and energy flow relative to frame u^{α} change by amounts of order ϵ:

$$\delta j_A{}^{\alpha} \equiv j_A{}^{\alpha}(\bar{u}) - j_A{}^{\alpha}(u) = -n_A\epsilon^{\alpha} + \epsilon O_1\,, \qquad \delta h^{\alpha} = -(\rho + P)\,\epsilon^{\alpha} + \epsilon O_1\,. \tag{59}$$

Equation (59) shows that there is a linear combination of $j_A{}^{\alpha}$ and h^{α} which is invariant to order ϵ. To appreciate the physical significance of this directly, define the (frame-invariant) *particle diffusion vector* $\nu_A{}^{\alpha}$ as the drift of species A seen in the energy frame $u_E{}^{\alpha}$:

$$\nu_A{}^{\alpha} \equiv j_A{}^{\alpha}(u_E) = j_A{}^{\alpha}(u) + n_A(u^{\alpha} - u_E{}^{\alpha}) + O_2 \tag{60}$$

(we have made use of (59) with $\bar{u} = u_E$.) Again from (59), since $h^{\alpha}(u_E) = 0$, we have

$$h^{\alpha}(u) = -(\rho + P)(u^{\alpha} - u_E{}^{\alpha}) + O_2 \tag{61}$$

and hence

$$\nu_A{}^{\alpha} = j_A{}^{\alpha}(u) - (\rho + P)^{-1}\, n_A h^{\alpha}(u) + O_2\,. \tag{62}$$

Since $j_A{}^{\alpha}(u_A) = 0$, (62) yields

$$\nu_A{}^{\alpha} = -(\rho + P)^{-1}\, n_A q_A{}^{\alpha} + O_2\,, \qquad q_A{}^{\alpha} \equiv h^{\alpha}(u_A). \tag{63}$$

The frame-invariant $q_A{}^\alpha$ is the energy flow measured by an observer who shares the mean particle velocity of species A. For a system of one component, this is what is conventionally understood by the term "heat flux."

6. Entropy

We now come to the central problem of irreversible thermodynamics: how to interrelate the primary variables S^μ, $N_A{}^\mu$, and $T^{\lambda\mu}$.

In general, for an arbitrary nonequilibrium state, no such relationship exists, as the example of a system describable by kinetic theory clearly shows. In kinetic theory, $N_A{}^\mu$ and $T^{\lambda\mu}$ are expressible in terms of first and second moments (in momentum space) of the particle distribution functions [5, 16]. Unless these functions are known a priori to have a special (local equilibrium or near equilibrium) form, two moments do not provide enough information to permit the enumeration of microscopic states needed to determine S^μ.

Thus any "hydrodynamical" theory, which attempts to characterize states of a fluid mixture by $N_A{}^\mu$ and $T^{\lambda\mu}$ only, is limited to near-equilibrium conditions.

Accepting this limitation, we shall assume that S^μ is a function of $N_A{}^\mu$ and $T^{\lambda\mu}$ for a fluid mixture satisfying the near-equilibrium conditions (46). In specifying the form of this relation, we shall be guided by the principle that it is to be strictly local, i.e., S^μ is to be independent of gradients of $N_A{}^\mu$ and $T^{\lambda\mu}$. This is certainly true in kinetic theory, where S^μ at any point is determined purely by the local form of the distribution functions as functions of momenta. More generally, it accords with the way one defines the entropy of an arbitrary system not in equilibrium [25]. One mentally divides the system into subsystems so small that their individual relaxation times are negligibly small in comparison with that of the system as a whole. Each subsystem can therefore be considered to be in a state of partial equilibrium. The entropy density at any point is then obtained by enumerating the microstates of all subsystems in a local element of volume which is macroscopically small but large enough to contain several subsystems. The resulting entropy density depends only on the local state, which is fully determined by the local values of $N_A{}^\mu$ and $T^{\lambda\mu}$ in a "hydrodynamical" theory.

It is convenient to write the relation sought in the form

$$S^\mu = [T(u)]^{-1}\,(P(u)\,u^\mu - u_\lambda T^{\lambda\mu}) - \sum_A \Theta_A(u)\,N_A{}^\mu - Q^\mu(u). \tag{64}$$

For the moment, this should be considered merely as a definition of $Q^\mu(u)$ in terms of the primary variables and an arbitrary frame velocity u^μ satisfying (48).

By virtue of (55), (56), and (31), the expression (64) can be reduced to

$$S^\mu = S(u)\, u^\mu + [T(u)]^{-1}\, h^\mu - \sum_A \Theta_A(u)\, j_A{}^\mu - Q^\mu(u). \tag{65}$$

Without Q^μ, this is precisely the linear relation between entropy flux, heat flux and particle diffusion which is imposed in the conventional quasi-stationary theory (e.g., for a simple fluid it reduces to $S^\mu = Su_N{}^\mu + T^{-1}q^\mu$ in Eckart's N-frame). It follows that Q^μ must be of at least second order in h^μ, $j_A{}^\mu$ to make the theory compatible with standard thermodynamics in the quasi-stationary limit. The most general such vector is (neglecting third-order terms and recalling that S^μ must be independent of gradients)

$$\begin{aligned} Q^\mu(u) &= \sigma \sum_A \alpha_0{}^A \nu_A{}^\mu + \sigma_\lambda{}^\mu \sum_A \alpha_1{}^A \nu_A{}^\lambda + \tfrac{1}{2} u^\mu \left(\beta_0 \sigma^2 + \sum_{A,B} \beta_1^{AB} \nu_A{}^\alpha \nu_{B\alpha} + \beta_2 \sigma_{\alpha\beta}\sigma^{\alpha\beta}\right) + R^\mu; \\ R^\mu &= T^{-1}(\rho + P)^{-1} (\tfrac{1}{2}\gamma_1 u^\mu h^\alpha h_\alpha + \gamma_2 \tau^{\mu\nu} h_\nu + \gamma_3 \sigma h^\mu)). \end{aligned} \tag{66}$$

The scalar coefficients $\alpha_i{}^A$, β_0, $\beta_1^{AB} = \beta_1^{AB}$, β_2, γ_i are undetermined functions of ρ and n_A.

The only part of the right side of (66) which is not frame-independent to order O_2 is R^μ. Under the change of frame (58), we obtain from (59),

$$\delta Q^\mu = Q^\mu(\bar{u}) - Q^\mu(u) = -T^{-1}(\gamma_1 u^\mu h^\nu + \gamma_2 \tau^{\mu\nu} + \gamma_3 \sigma g^{\mu\nu})\, \epsilon_\nu + \epsilon O_2\,. \tag{67}$$

This means that the coefficients γ_i are not arbitrary, but must be chosen to give the correct law of transformation of Q^μ under a change of frame. This law is easily inferred from the defining equation (64). We of course have $\delta N_A{}^\mu = \delta T^{\lambda\mu} = \delta S^\mu = 0$, and, recalling that changes of thermodynamic parameters are of order ϵO_1,

$$\begin{aligned} &\delta(P/T)\, u^\mu - \delta(T^{-1})\, u_\lambda T^{\lambda\mu} - \sum_A N_A{}^\mu\, \delta\Theta_A \\ &\qquad = \left[\delta(P/T) + \rho\, \delta(T^{-1}) - \sum_A n_A\, \delta\Theta_A\right] u^\mu + \epsilon O_2\,. \end{aligned} \tag{68}$$

The expression in square brackets vanishes to order ϵO_1 (linear in ϵ) because of (36). We thus obtain

$$T\delta Q^\mu = (Pg^{\mu\nu} - T^{\mu\nu})\, \delta u_\nu + \epsilon O_2 + O(\epsilon^2) = -(u^\mu h^\nu + \tau^{\mu\nu})\, \epsilon_\nu + \epsilon O_2 + O(\epsilon^2). \tag{69}$$

Equating coefficients in (67) and (69) for the case $\epsilon \ll O_1$ yields

$$\gamma_1 = \gamma_2 = 1, \qquad \gamma_3 = 0. \tag{70}$$

By combining (64), (66), and (70) we obtain the complete expression for S^μ as a function of $N_A{}^\mu$ and $T^{\lambda\mu}$.

7. Entropy Production and Phenomenological Laws

The divergence of (64) gives, for the entropy production

$$S^{\mu}{}_{|\mu} = -T^{-1}u_{\lambda}T^{\lambda\mu}{}_{|\mu} - \sum_{A} \Theta_A N^{\mu}_{A|\mu} - Q^{\mu}{}_{|\mu}$$

$$+ (T^{-1}Pu^{\mu})_{|\mu} - (T^{-1}u_{\lambda})_{|\mu}\, T^{\lambda\mu} - \sum_{A} \Theta_{A,\mu} N_A{}^{\mu}. \tag{71}$$

If we substitute for $T^{\lambda\mu}$ and $N_A{}^{\mu}$ from (55) and (56), the last three terms (second line) become

$$u^{\mu}\left\{(T^{-1}P)_{,\mu} + \rho(T^{-1})_{,\mu} - \sum_{A} \Theta_{A,\mu} n_A\right\} - (T^{-1}u_{\lambda})_{|\mu}\,(2h^{(\lambda}u^{\mu)} + \tau^{\lambda\mu}) - \sum_{A} \Theta_{A,\mu} j_A{}^{\mu},$$

and the expression in braces vanishes according to the identity (36). Hence

$$S^{\mu}{}_{|\mu} = -T^{-1}u_{\lambda}T^{\lambda\mu}{}_{|\mu} - \sum_{A} \Theta_A N^{\mu}_{A|\mu} + W - Q^{\mu}{}_{|\mu}\,, \tag{72}$$

where

$$W = h^{\mu}[(T^{-1})_{,\mu} - T^{-1}\dot{u}_{\mu}] - T^{-1}\tau^{\lambda\mu}u_{\lambda|\mu} - \sum_{A} j_A{}^{\mu}\Theta_{A,\mu}\,. \tag{73}$$

(The term involving $T^{\lambda\mu}{}_{|\mu}$ of course vanishes in the case of interest to us; it has been explicitly retained in (72) for application to non-isolated systems in the following section.)

Those terms of (72) which are not frame-invariant—namely, W and $-R^{\mu}{}_{|\mu}$—must nevertheless have a frame-invariant sum, since the remaining terms are manifestly invariant to order O_2. The sum can be expressed in a manifestly frame-invariant form:

$$W - R^{\mu}{}_{|\mu} = -\sum_{A} \nu_A{}^{\mu}\Theta_{A,\mu} - T^{-1}\tau^{\lambda\mu}(u_{\lambda} + (\rho + P)^{-1}\, h_{\lambda})_{|\mu} + O_3\,. \tag{74}$$

Since both sides of (74) are frame-invariant (recall that

$$u_{\lambda} + (\rho + P)^{-1}\, h_{\lambda} = u_{\lambda}{}^{E} + O_2 \tag{75}$$

from (61)) (74) is immediately verified by specializing to the E-frame, where $h^{\mu} = 0$, $j_A{}^{\mu} = \nu_A{}^{\mu}$. Alternatively, one can work in a general frame, using a result analogous to (16).

Recalling (25), we can write

$$\sum_{A} \Theta_A N^{\mu}_{A|\mu} = T^{-1} \sum_{i} \mathcal{A}_i r_i \tag{76}$$

where

$$\mathcal{A}_i = T \sum_A \Theta_A c_{Ai} \tag{77}$$

is the "chemical affinity" of the ith reaction; in equilibrium it vanishes according to (32).

In calculating $Q^\mu{}_{|\mu}$ from (66), we may treat $u^\mu{}_{|\mu}$ and the time-derivatives of the coefficients $\beta(\rho, n_A)$ as O_1 quantities, since they vanish for equilibrium by (38) and (39).

Substituting into (72) from (66), (74) and (76), we obtain the entropy production

$$\begin{aligned} S^\mu{}_{|\mu} = &- T^{-1} \sum_i \mathcal{A}_i r_i - \sum_A \nu_A{}^\mu \left(\Theta_{A,\mu} + \sum_B \beta_1^{AB} \dot{\nu}_{B\mu} + \alpha_0{}^A \sigma_{,\mu} + \alpha_1{}^A \sigma^\nu_{\mu|\nu} \right) \\ &- \sigma \left(T^{-1}(u^\alpha + (\rho + P)^{-1} h^\alpha)_{|\alpha} + \beta_0 \dot{\sigma} + \sum_A (\alpha_0{}^A \nu_A{}^\mu)_{|\mu} \right) \\ &- \sigma^{\mu\nu} \left(T^{-1}(u_\mu + (\rho + P)^{-1} h_\mu)_{|\nu} + \beta_2 \dot{\sigma}_{\mu\nu} + \sum_A (\alpha_1{}^A \nu_{A\mu})_{|\nu} \right) + O_3 . \end{aligned} \tag{78}$$

In extracting the phenomenological laws from (78), it is important to note that the Curie principle does not hold in general, since the equilibrium state of a rotating, self-gravitating fluid singles out two preferred spatial directions: the local axis of rotation and the direction of the local gravitational field (i.e., pressure gradient). This admits a host of couplings between the viscous stresses and heat flow, some curious (e.g., circular heat flow in a differentially rotating fluid) and some of practical importance (e.g., "radiative viscosity" [17]). Since most of these effects are not "new" (they are already present at the quasi-stationary level), we shall not trouble to write out the full panoply of coupling terms. Let us restrict ourselves to the case where the angular velocity and pressure gradient in equilibrium are sufficiently small that we can neglect their products with O_1 terms in the phenomenological laws. This implies in particular that we can neglect gradients of $(\rho + P)$ and $\alpha_i{}^A(\rho, n_A)$ in (78).

Assuming that the rates and stresses r_i, σ, $\sigma^{\mu\nu}$, $\nu_A{}^\mu$ depend linearly on the "strains" $\mathcal{A}_i$, $u^\alpha{}_{|\alpha}$, $\langle u_{\mu|\nu} \rangle$, $\Theta_{A,\mu}$, we then infer from $S^\mu{}_{|\mu} \geqslant 0$ that, to O_1,

$$r_i = \sum_j \Lambda_{ij} \mathcal{A}_j + \Lambda_i (u^\alpha{}_{|\alpha} + (\rho + P)^{-1} h^\alpha{}_{|\alpha}) \tag{79}$$

$$\nu_A{}^\alpha = -\Delta^{\alpha\beta} T^2 \sum_B \kappa_{AB} \left(\Theta_{B,\beta} + \sum_C \beta_1^{BC} \dot{\nu}_{C\beta} + \alpha_0{}^B \sigma_{,\beta} + \alpha_1{}^B \sigma^\nu_{\beta|\nu} \right) \tag{80}$$

$$\sigma = -\tfrac{1}{3}\zeta_V\left(u^{\alpha}{}_{|\alpha} + (\rho + P)^{-1}\overset{\alpha}{h}{}_{|\alpha} + T\beta_0\dot{\sigma} + T\sum_A \alpha_0{}^A\nu^{\mu}_{A|\mu}\right) - \sum_i \Lambda_i\mathcal{A}_i \quad (81)$$

$$\sigma_{\alpha\beta} = -2\zeta\left\langle u_{\alpha|\beta} + (\rho + P)^{-1}h_{\alpha|\beta} + T\beta_2\dot{\sigma}_{\alpha\beta} + T\sum_A \alpha_1{}^A\nu_{A\alpha|\beta}\right\rangle. \quad (82)$$

To ensure positivity of $S^{\mu}{}_{|\mu}$ for arbitrary "strains," the shear and bulk viscosities ζ, ζ_V must be nonnegative, and the reaction and diffusion matrices Λ_{ij}, κ_{AB} positive semidefinite.

Onsager's principle of reciprocity requires symmetry of the matrices Λ_{ij}, κ_{AB}, and $\sum_B \kappa_{AB}\beta_1^{BC}$. Since β_1^{BC} is symmetric by definition, it follows that κ_{AB} and β_1^{BC} commute.

Because equilibrium is the stable state of an isolated system and maximizes the entropy, transient transport processes generated (at given ρ, n_A) in a system initially in equilibrium should *diminish* the entropy density ($-u_\mu S^\mu$). According to (65) and (66), this implies that β_0, β_2 are nonnegative, and the matrix β_1^{AB} is positive semidefinite. Hence $\sum_B \kappa_{AB}\beta_1^{BC}$ is also positive semidefinite. Thus all relaxation times are essentially positive.

Equations (79)–(82) are the main results of the paper and generalize the quasi-stationary phenomenological laws for mixtures as given, for example by Aitken [3] and Stewart [5]. The form which they assume for simple fluids has been discussed in Section 2.

Equation (80) amalgamates [3] Fick's law and Fourier's law by referring particle drifts to the mean energy flow. An explicit heat flux q^μ can be introduced, if desired, by choosing a "material 4-velocity"

$$u_M{}^\mu = \left(\sum_A W_A N_A{}^\mu\right)\Big/\left(\sum_A W_A n_A\right) = u_E{}^\mu + \left(\sum_A W_A \nu_A{}^\mu\right)\Big/\left(\sum_A W_A n_A\right) \quad (84)$$

where W_A are arbitrary weights (e.g., rest-mass or baryon number for a particle of species A), and then defining q^μ as the energy flux observed in the "material rest-frame":

$$q^\mu = h^\mu(u_M) = -(\rho + P)\left(\sum_A W_A \nu_A{}^\mu\right)\Big/\left(\sum_A W_A n_A\right) \quad (85)$$

In writing (84) and (85), we have called on (60) and (61).

8. Heat Exchange Between Rapidly Moving Bodies

We now examine a method of thermodynamical description differing from that of the preceding sections in that it is based on the assumption of weak interactions between the components rather than their closeness to a common equi-

librium. This will enable us to dispense with the condition (47), which required the relative velocities of components to be small.

We assume each component to be nearly homogeneous, close to an individual ("partial") equilibrium, and described by an individual energy tensor $T_A^{\lambda\mu}$, number flux $N_A{}^\mu$ and entropy flux $S_A{}^\mu$ whose divergences are small. This mode of description is actually quite general, and applies to an arbitrary system if we define the "components" to be subsystems having a sufficiently small spatial extension.

For simplicity, we shall idealize the situation further by assuming the internal relaxation times of the components to be so small that the partial energy tensors and fluxes can be taken to have their equilibrium form:

$$T_A^{\lambda\mu} = \rho_A u_A{}^\lambda u_A{}^\mu + P_A \Delta_A^{\lambda\mu}, \qquad N_A{}^\mu = n_A u_A{}^\mu \tag{86}$$

$$S_A{}^\mu = T_A^{-1}(P_A u_A{}^\mu - u_{A\lambda} T_A^{\lambda\mu}) - \Theta_A N_A{}^\mu = S_A u_A{}^\mu. \tag{87}$$

The total energy tensor $T^{\lambda\mu} = \sum_A T_A^{\lambda\mu}$ and the total entropy flux $S^\mu = \sum_A S_A{}^\mu$ satisfy

$$T^{\lambda\mu}{}_{|\mu} = 0, \qquad S^\mu{}_{|\mu} \geqslant 0. \tag{88}$$

A simple calculation gives (cf. Eq. (72))

$$S^\mu_{A|\mu} = -\Theta_A N^\mu_{A|\mu} - T_A^{-1} u_{A\lambda} T_A^{\lambda\mu}{}_{|\mu}, \tag{89}$$

so that

$$S^\mu{}_{|\mu} = -\sum_A \Theta_A N^\mu_{A|\mu} - \sum_A T_A^{-1} u_{A\lambda} T_A^{\lambda\mu}{}_{|\mu} \geqslant 0. \tag{90}$$

Assume now that the linear dimensions of each component are small compared to the radius of curvature of space–time, and that T_A, Θ_A, and $u_A{}^\lambda$ are effectively constant over A. The increment of A's 4-momentum in a given short interval is then

$$\begin{aligned} \delta P_A{}^\mu &= \delta t \frac{\partial}{\partial t} \int T_A^{\mu 4} d^3x, \\ &= \delta t \int T_A^{\mu\nu}{}_{|\nu} d^3x \end{aligned} \tag{91}$$

(in local geodesic coordinates with arbitrary time-axis). Similarly, the incremenst of particle number and entropy are

$$\delta n_A = \delta t \int N^\mu_{A|\mu} d^3x, \qquad \delta S = \delta t \int S^\mu{}_{|\mu} d^3x. \tag{92}$$

From (90) we can now infer the *discrete form of the second law of thermodynamics*

$$\delta S = -\sum_A \Theta_A \, \delta n_A - \sum_A T_A^{-1} u_{A\lambda} \, \delta P_A{}^\lambda \geqslant 0 \tag{93}$$

for the exchange of 4-momentum and matter between a set of bodies in arbitrary relative motion. In special relativity these exchanges are subject to the overall conservation law

$$\sum_A \delta P_A{}^\mu = 0 \tag{94}$$

which follows by integration of (88). In general relativity, addition of vectors at different locations is not well defined, and (94) only has a meaning if the spatial separation of the interacting bodies is small compared to the radius of curvature. Physically, the breakdown of (94) for wide separations is, of course, associated with work done against the gravitational forces in the transfer of $\delta P_A{}^\mu$. In any event, (93) is valid quite generally.

Although our derivation has only been concerned with fluids, it should be emphasized that the final form (93) of the second law applies to exchanges between arbitrary non-fluid bodies. This is because we can imagine the nonfluid bodies handing over the transferred 4-momentum and particles—reversibly—to neighboring fluid bodies with the same $u_A{}^\alpha$, T_A, Θ_A, then letting the irreversible exchanges occur between the fluid bodies.

Equation (93) shows, for instance, that if the center of a spoked wheel, rotating with rim speed v and rim temperature T_0, is cooled to temperature $T_1 < T_0$, heat will still flow radially outward to the warmer rim, provided $T_0(1 - v^2)^{1/2} < T_1$ (cf. Eq. (45)). Questions of this kind have received some attention in the literature [18]. The form (93) of the second law (for exchange of 4-momentum only) seems to have been first given by van Kampen [19].

9. Concluding Remarks

It is reasonable to expect that signal speeds of thermal disturbances will be of the order of the speed of sound if the relaxation times in the phenomenological equations are of the order of the mean collision time. Detailed studies which confirm both of these expectations have been carried out by Kranyš [20] who has derived equations similar to (8) (although neglecting bulk viscosity) on the basis of the Grad approximation in relativistic kinetic theory. Similar results have been announced by Stewart [21]. (We hope to return to this subject in a future publication.) Within their range of validity the kinetic-theoretical derivations of course yield much more information, but the phenomenological theory is still

needed to lend credence to the results by showing that they are not artifacts of a particular model or approximation.

This investigation was prompted in part by a remark by Carter [22] that a finite speed of heat propagation could be obtained by adding an ad hoc term $-(cT)^{-2}\, q_\alpha d(S^{-1}q^\alpha)$ to the usual form (30) of the Gibbs relation. After completion of the work I became aware of a paper by Müller [23], which presents similar considerations for a nonrelativistic simple fluid, and mentions that a relativistic formulation is to be found in Müller's thesis (see also Müller [24] for an attempt at a relativistic theory).

Appendix: Effect of a Change of Rest-Frame

To verify statements (i) and (ii) of Section 5, it is convenient to start from the decompositions of $T^{\alpha\beta}$ and $N_A{}^\alpha$ in the energy frame:

$$T^{\alpha\beta} = (\rho + P)_E\, u_E{}^\alpha u_E{}^\beta + P_E g^{\alpha\beta} + \tau_E^{\alpha\beta}, \qquad \tau_E^{\alpha\beta} u_{E\beta} = 0 \tag{95}$$

$$N_A{}^\alpha = n_A(u_E)\, u_E{}^\alpha + \nu_A{}^\alpha, \qquad \nu_A{}^\alpha u_{E\alpha} = 0. \tag{96}$$

If u^α is an arbitrary unit timelike vector, we can set

$$u_E{}^\alpha = (1 + \delta^2)^{1/2}\, u^\alpha + \delta^\alpha, \qquad \delta^\alpha u_\alpha = 0, \qquad \delta^2 = \delta^\alpha \delta_\alpha\,. \tag{97}$$

We assume $\delta = O_1$ and neglect terms of order O_3. Then

$$\Delta_\beta{}^\alpha(u)\, u_E{}^\beta = \delta^\alpha, \qquad u_\beta \Delta_E^{\alpha\beta} = -\delta^\alpha - \delta^2 u^\alpha, \qquad \Delta_\beta{}^\alpha(u)\, \Delta_E^{\beta\gamma} = \Delta_E^{\alpha\gamma} - u^\alpha \delta^\gamma \tag{98}$$

$$\rho(u) = u_\alpha u_\beta T^{\alpha\beta} = \rho_E + (\rho + P)_E\, \delta^2 \tag{99}$$

$$n_A(u) = -u_\alpha N_{A\alpha} = n_A(u_E)(1 + \delta^2)^{1/2} + \nu_A{}^\alpha \delta_\alpha \tag{100}$$

$$h^\mu(u) = -\Delta_\alpha{}^\mu(u)\, u_\beta T^{\alpha\beta} = (\rho + P)_E\, \delta^\mu + \tau_E^{\mu\beta} \delta_\beta \tag{101}$$

$$j_A{}^\mu(u) = \Delta_\alpha{}^\mu(u)\, N_A{}^\alpha = \nu_A{}^\mu + n_A \delta^\mu - u^\mu \nu_A{}^\beta \delta_\beta \tag{102}$$

$$\tau^{\mu\nu}(u) = \Delta_\alpha{}^\mu(u)\, \Delta_\beta{}^\nu(u)\, T^{\alpha\beta} - P(u)\, \Delta^{\alpha\beta} = \tau_E^{\mu\nu} + (\rho + P)_E\, \delta^\mu \delta^\nu + [P_E - P(u)]\, \Delta^{\mu\nu}. \tag{103}$$

From (30),

$$S(u) - S(u_E) = T_E^{-1}(\rho(u) - \rho_E) - \sum_A \Theta_A(u_E)(n_A(u) - n_A(u_E)) \tag{104}$$

and changes in the other thermodynamic variables may be obtained similarly; all are of order O_2.

The effects of the transformation (58) are now easily read off from the preceding formulas by setting $\epsilon^\alpha = \delta^\alpha - \bar{\delta}^\alpha$.

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Note added in proof. Explicit values for the coefficients α_i, β_i in Eqs. (8) have now been obtained [26] for simple relativistic Boltzmann, Bose, and Fermi gases, using the Grad 14-moment approximation in relativistic kinetic theory. It may be useful to record here the results for a Boltzmann gas.

Let $\beta = m/kT$, $\bar{\eta} = \eta/m = (\rho + P)/nm$, where m is the particle rest mass. Then [16]

$$\bar{\eta} = K_3(\beta)/K_2(\beta), \qquad \gamma/(\gamma - 1) = \beta^2(1 - \bar{\eta}^2 + 5\bar{\eta}/\beta),$$

where $K_n(\beta)$ are modified Bessel functions, and γ is the ratio of specific heats. Further, define

$$\Omega = 3\gamma - 5 + /(\bar{\eta}\beta), \qquad \Omega^* = 5 - 3\gamma + 3(10 - 7\gamma)\bar{\eta}/\beta,$$
$$\Omega^{**} = 5 - 3\gamma - 3\gamma^2/[(\gamma - 1)\eta^2\beta^2].$$

Then

$$\alpha_0 = (\gamma - 1)\Omega^{**}/(\gamma P\Omega), \qquad \alpha_1 = -(\gamma - 1)/(\gamma P),$$
$$\beta_0 = 3\Omega^*/(\bar{\eta}^2 P\Omega^2), \qquad \beta_1 = [(\gamma - 1)/\gamma]^2\beta(\bar{\eta}P)^{-1}[5\eta^2 - \gamma/(\gamma - 1)],$$
$$\beta_2 = (2\bar{\eta}^2 P)^{-1}(1 + 6\bar{\eta}/\beta).$$

In the nonrelativistic limit ($\beta \to \infty$)

$$\alpha_0 \approx \tfrac{4}{5}\beta P^{-1}, \qquad \alpha_1 \approx -\tfrac{2}{5}P^{-1}, \qquad \beta_0 \approx \tfrac{6}{5}\beta^2 P^{-1},$$
$$\beta_1 \approx \tfrac{2}{5}\beta P^{-1}, \qquad \beta_2 \approx \tfrac{1}{2}P^{-1}.$$

In the photon limit ($\beta \to 0$)

$$\alpha_0 \approx \tfrac{1}{2}P^{-1}, \qquad \alpha_1 \approx -\tfrac{1}{4}P^{-1}, \qquad \beta_0 \approx 216\beta^{-4}P^{-1},$$
$$\beta_1 \approx \tfrac{5}{4}P^{-1}, \qquad \beta_2 \approx \tfrac{3}{4}P^{-1}.$$

Generally, one finds for gases that the coefficients α_i, β_i are independent of the details of interaction between the particles (collision cross-sections, etc.). This implies that the wave-front speeds of thermal and viscous disturbances (like the speed of sound) depend only on the thermodynamical state n, T; for a Boltzmann gas they are functions of T only. Detailed analysis [26] shows that all propagation speeds are of the order of the speed of sound. The fastest mode approaches an upper limit of $[(18 + (69)^{1/2})/45]^{1/2}c = 0.765c$ as T.

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595/100/1-2-22